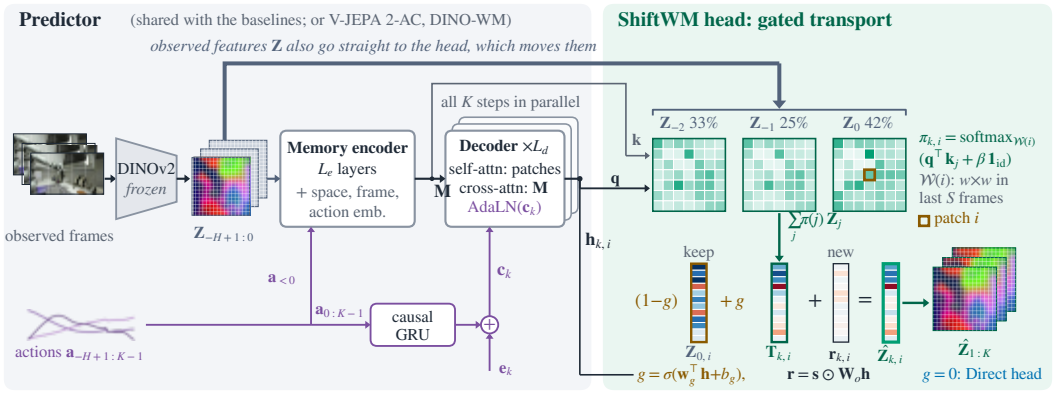

Predictor
(shared with the baselines; or V-JEPA 2-AC, DINO-WM)
observed features Z also go straight to the head, which moves them
ShiftWM head: gated transport
observed frames
DINOv2
frozen
Z−H+1:0
Memory encoder
Le layers
+ space, frame, action emb.
M
all K steps in parallel
Decoder ×Ld
self-attn: patches
cross-attn: M
AdaLN(ck)
q
k
Z−2 33%
Z−1 25%
Z0 42%
πk,i = softmaxW(i)
(qᵀ kj + β 1id)
W(i): w×w in last S frames
patch i
∑j π(j) Zj
hk,i
a<0
a0:K−1
ck
causal GRU
ek
actions a−H+1:K−1
keep
new
(1−g)
+ g
+
=
Z0,i
Tk,i
rk,i
Ẑk,i
Ẑ1:K
g = σ(wgᵀ h + bg),
r = s ⊙ Wo h
g = 0: Direct head